

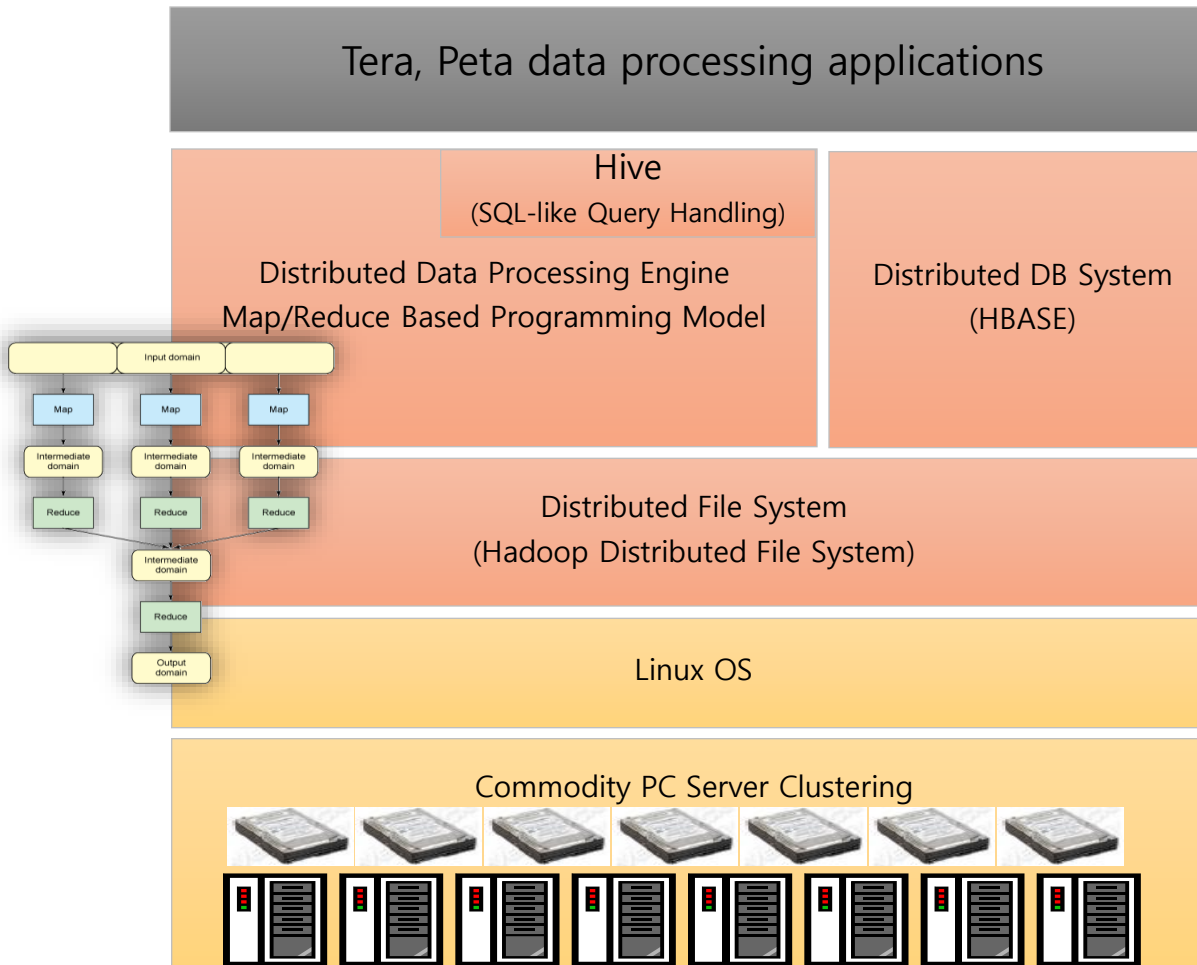
Introduction

Big Data Analytics

Hadoop

What is Hadoop?

Hadoop is a Open Source Platform based on Java. Hadoop has HDFS(hadoop Distributed File System) which is similar to GFS. Based on HDFS, Hadoop provides Map/Reduce Job execution on a cluster composed of thousands of PCs.



[Apache Hadoop Project]

◆ Hadoop Core

- Distributed File System
- Map/Reduce Framework

◆ Pig (initiated by Yahoo!)

- Parallel Programming Language and Runtime

◆ Hbase (initiated by Powerset)

- Table storage for semi-structured data

◆ Hive (initiated by Facebook)

- SQL-like query language & meta-store

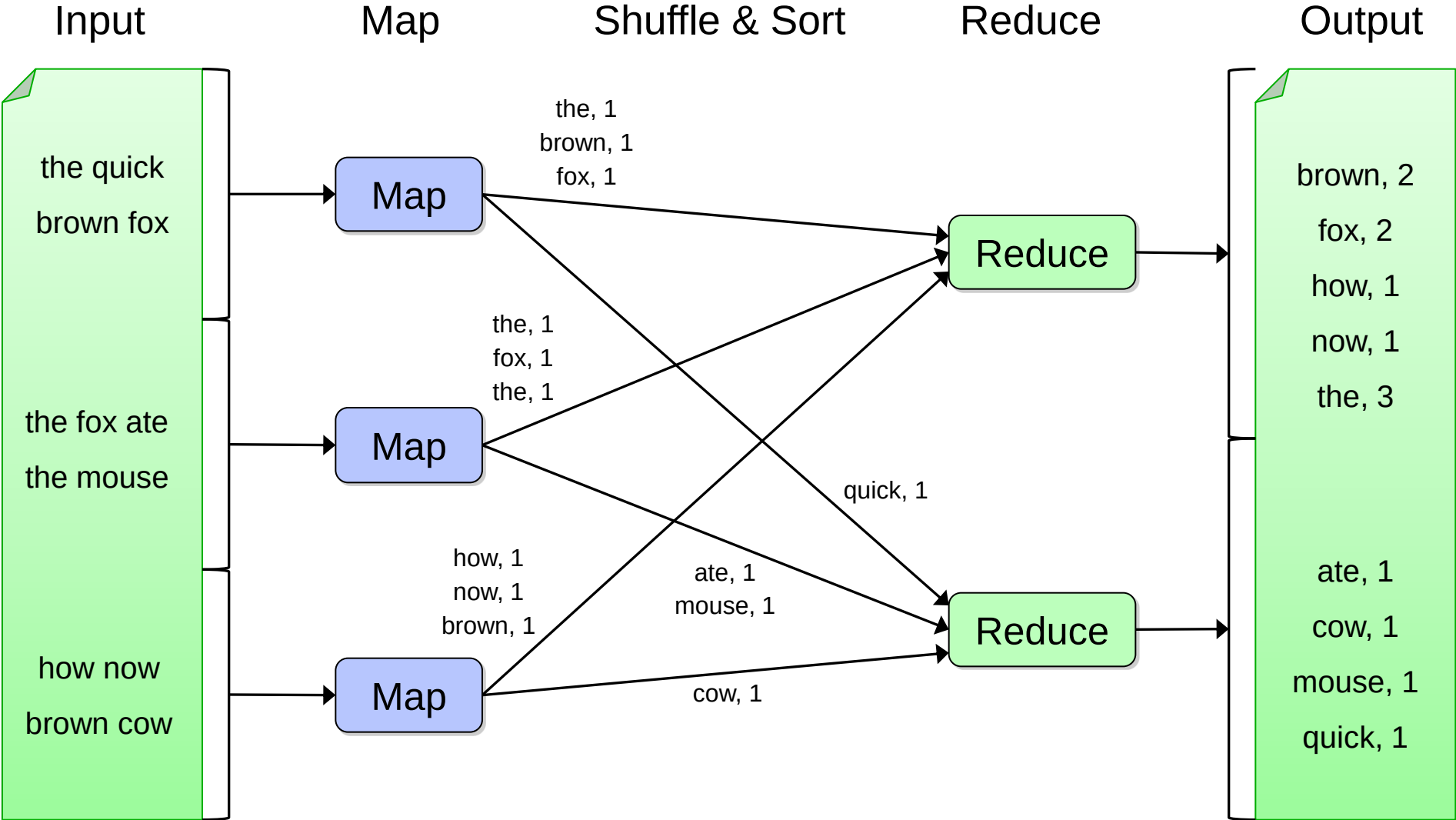
Redhat, CentOS, Ubuntu ...

◆ PC Server

- CPU :Intel Quad Core x 2 CPU
- Memory: 16GB
- HDD: 500GB x 4
- NIC: Gigabit Ethernet x 2 Port

Map-Reduce

Word Count example



```
public class MapClass extends MapReduceBase
    implements Mapper<LongWritable, Text, Text, IntWritable> {

    private final static IntWritable ONE = new IntWritable(1);

    public void map(LongWritable key, Text value,
                    OutputCollector<Text, IntWritable> out,
                    Reporter reporter) throws IOException {
        String line = value.toString();
        StringTokenizer itr = new StringTokenizer(line);
        while (itr.hasMoreTokens()) {
            out.collect(new text(itr.nextToken()), ONE);
        }
    }
}
```

```
public class ReduceClass extends MapReduceBase
    implements Reducer<Text, IntWritable, Text, IntWritable> {

    public void reduce(Text key, Iterator<IntWritable> values,
                        OutputCollector<Text, IntWritable> out,
                        Reporter reporter) throws IOException {
        int sum = 0;
        while (values.hasNext()) {
            sum += values.next().get();
        }
        out.collect(key, new IntWritable(sum));
    }
}
```

```
public static void main(String[] args) throws Exception {  
    JobConf conf = new JobConf(WordCount.class);  
    conf.setJobName("wordcount");  
  
    conf.setMapperClass(MapClass.class);  
    conf.setCombinerClass(ReduceClass.class);  
    conf.setReducerClass(ReduceClass.class);  
  
    FileInputFormat.setInputPaths(conf, args[0]);  
    FileOutputFormat.setOutputPath(conf, new Path(args[1]));  
  
    conf.setOutputKeyClass(Text.class); // out keys are words (strings)  
    conf.setOutputValueClass(IntWritable.class); // values are counts  
  
    JobClient.runJob(conf);  
}
```

Contents

1. Cloud Computing
2. Map-Reduce
3. Hadoop
4. Big Data Analysis
 - Motivation
 - Possible approaches
 - Collaborative Filtering
 - CF Technologies
 - Sample Projects

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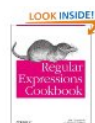


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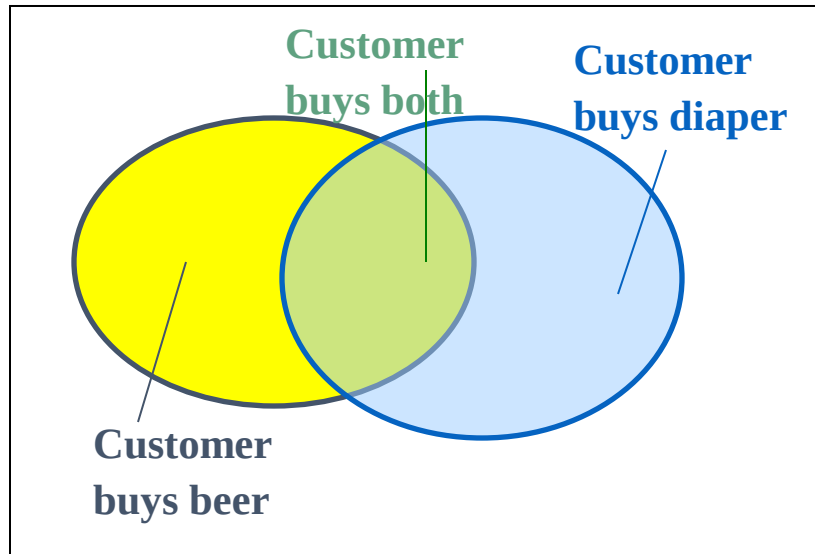
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*Book Recommendation by
Collaborative Filtering*

Basic Concepts: Frequent Patterns

Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- **itemset**: A set of one or more items
- **k-itemset** $X = \{x_1, \dots, x_k\}$
- **(absolute) support**, or, **support count** of X : Frequency or occurrence of an itemset X
- **(relative) support**, s , is the fraction of transactions that contains X (i.e., the probability that a transaction contains X)
- An itemset X is **frequent** if X 's support is no less than a *minsup* threshold

When Harry Met Sally (1989)



Add to DVD Queue



Average of raters like you: **4.6** stars
Average of 1,865,680 ratings: **4.0** stars

DETAILS

Available Formats

DVD

 Restricted. Under 17 requires accompanying parent or adult guardian.

AGE 15

Iffy for 15+ [\(more\)](#)

At A Glance

Friends

Member Reviews

Critics

More Like This

 **Play the Preview**

Can men and women remain friends without sex getting in the way? Nora Ephron's episodic screenplay introduces womanizing, neurotic Harry (Billy Crystal) and ambitious, equally neurotic Sally (Meg Ryan) as chums who resist sexual attraction to maintain their friendship – a relationship always teetering on the brink of love. As the two draw closer, the question resurfaces: Can they stay just pals? Carrie Fisher and Bruno Kirby co-star.

ENJOYED BY MEMBERS WHO ENJOYED

Bull Durham

On Golden Pond

The Big Chill

Roxanne

Sex and the City: Season 4 (3-Disc Series)

Movie Recommendation by Collaborative Filtering



CRITIC REVIEWS

 [Roger Ebert, Chicago Sun-Times](#)

What makes it special, apart from the Faber screenplay, is the character between

Customer0: {0, 1, 0, 1}

Customer1: {1, 0, 1, 0}

Customer2: {1, 1, 0, 0}

Customer3: {1, 1, 0, 1}

$$\text{Jaccard}(C_i, C_j) = \frac{|C_i \cap C_j|}{|C_i \cup C_j|}$$

The screenshot shows the Netflix homepage for user Soojin Kim. The top navigation bar includes links for 'Browse DVDs', 'Watch Instantly', 'Your Queue', 'Movies You'll ♥', and 'Instantly to your TV'. A search bar is located on the right. Below the navigation bar, there are links for 'Home', 'Genres', 'New Releases', 'Friends', 'Netflix Top 100', 'Critics' Picks', and 'Award Winners'.

The main content area features a section titled 'Because you enjoyed:' with a list of movies: 'Billy Elliot', 'The Pianist', and 'Gandhi'. Below this, it says 'We think you'll enjoy:' followed by 'Life Is Beautiful'. A blue oval highlights the 'Add' button and the text 'Movie Recommendation by Collaborative Filtering'.

To the right of the 'Because you enjoyed:' section is a 'YOUR RECENT ACTIVITY' box showing two entries: '10/02 We received Lost: Season 1: Disc 2' and '10/01 We received Lost: Season 1: Disc 1'.

Below the main content area is a section titled 'Independent Comedies Featuring a Strong Female Lead' with a 'See all >' link. This section displays a carousel of movie posters: 'Kissing Jessica Stein', 'Calendar Girls', 'Then She Found Me', 'The Ramen Girl', 'Gray Matters', and 'Suburban Girl'.

Customer1: {0, 1, 0, 1}

Customer2: {1, 0, 1, 0}

Customer3: {1, 1, 0, 0}

Customer4: {1, 1, 0, 1}

Movie1: {0, 1, 1, 1}

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Movie3: {0, 1, 0, 0}

Movie4: {1, 0, 0, 1}

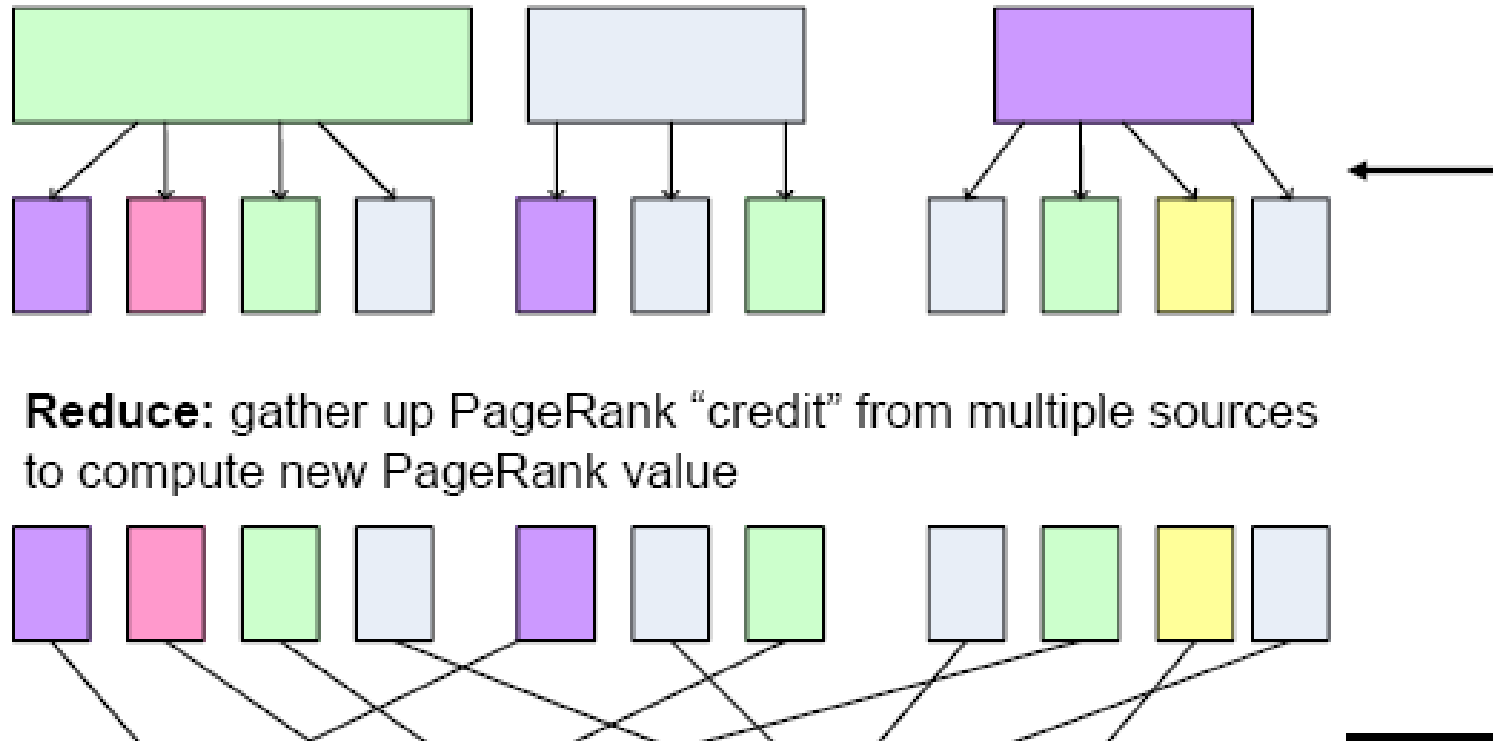
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Churn Management

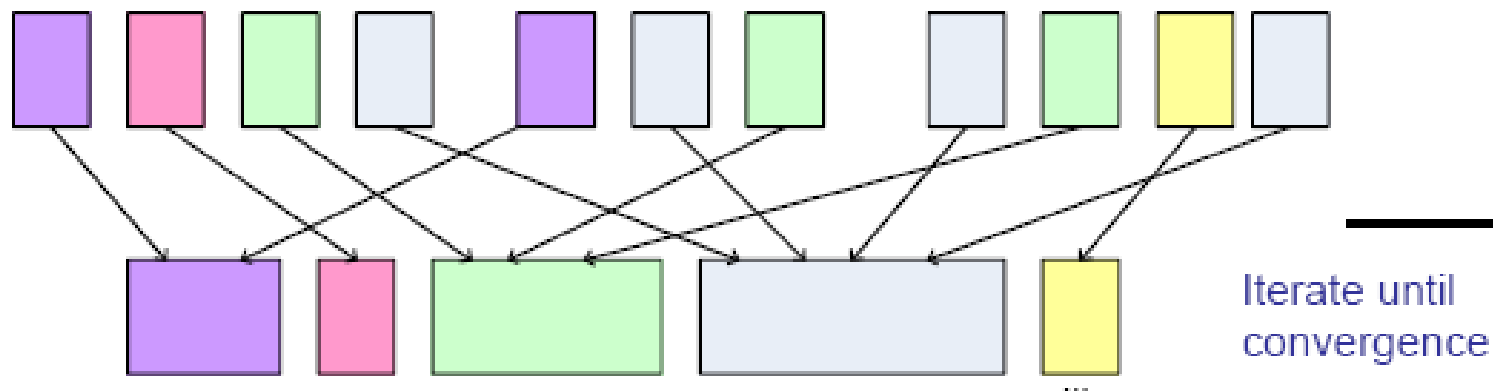
Ad

**{"S_id": "xx", "R_id": "yy", "S_time": "20201010150021", "E_time": "20201010150313",
"S_cid": "aa", "R_cid": "bb"}**

Map: distribute PageRank “credit” to link targets



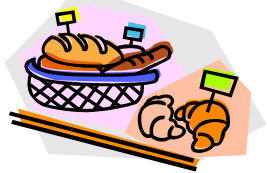
Reduce: gather up PageRank “credit” from multiple sources to compute new PageRank value



Advertisement

1 Prepare Vouchers

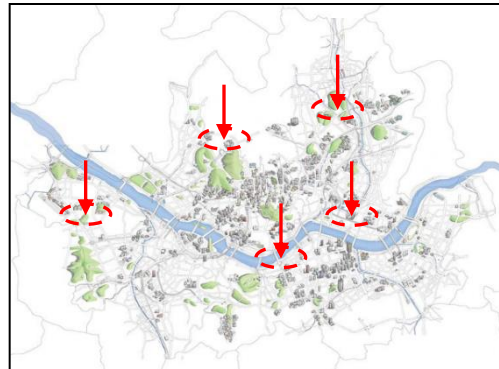
Vouchers for Breads



Vouchers for Coffee



2 Select 5 Spots in the City



3 Send Vouchers to TWO groups

1st
Group

Destination Based Approach

- Vouchers were sent to the people who drives to one of 5 locations

2nd
Group

Segment Based Approach

- Vouchers were sent to the Segment : 'who lives in or frequently visits to' one of five locations

Results
&
Insights

✓ Second group showed 3x better response rate than first group

- What are these for?



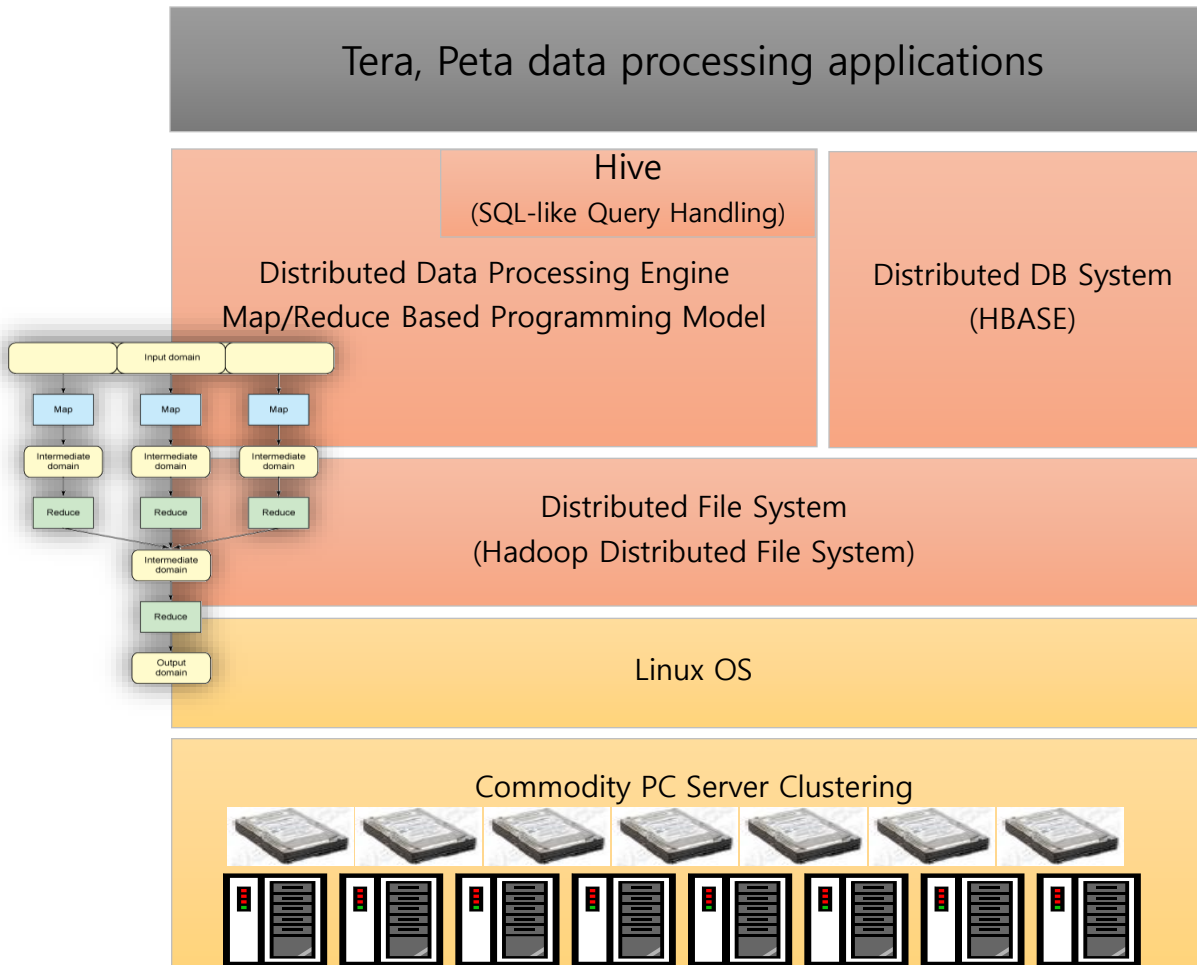
Hadoop

Big Data Analytics

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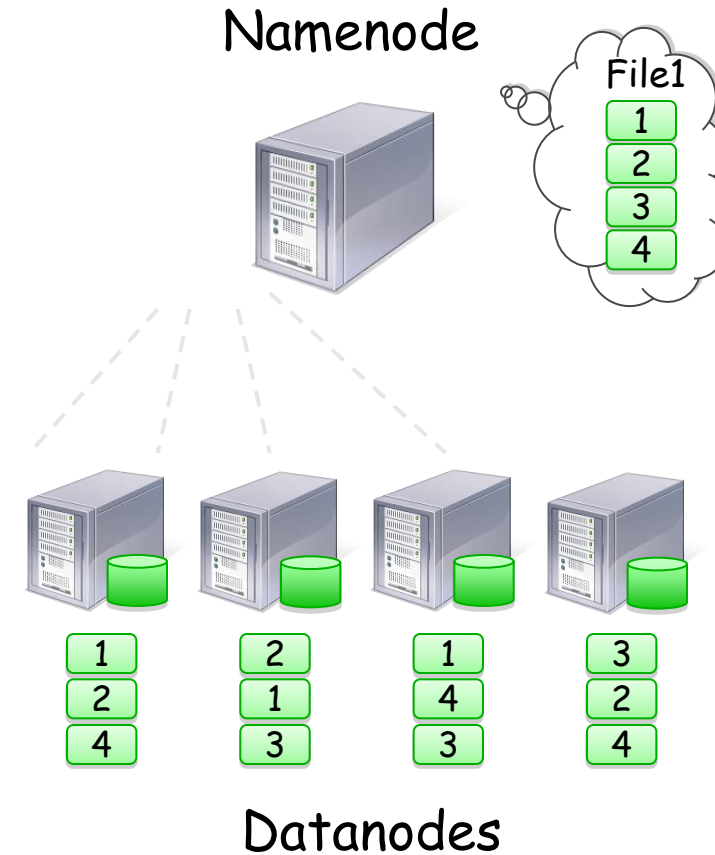
- SQL-like query language & meta-store

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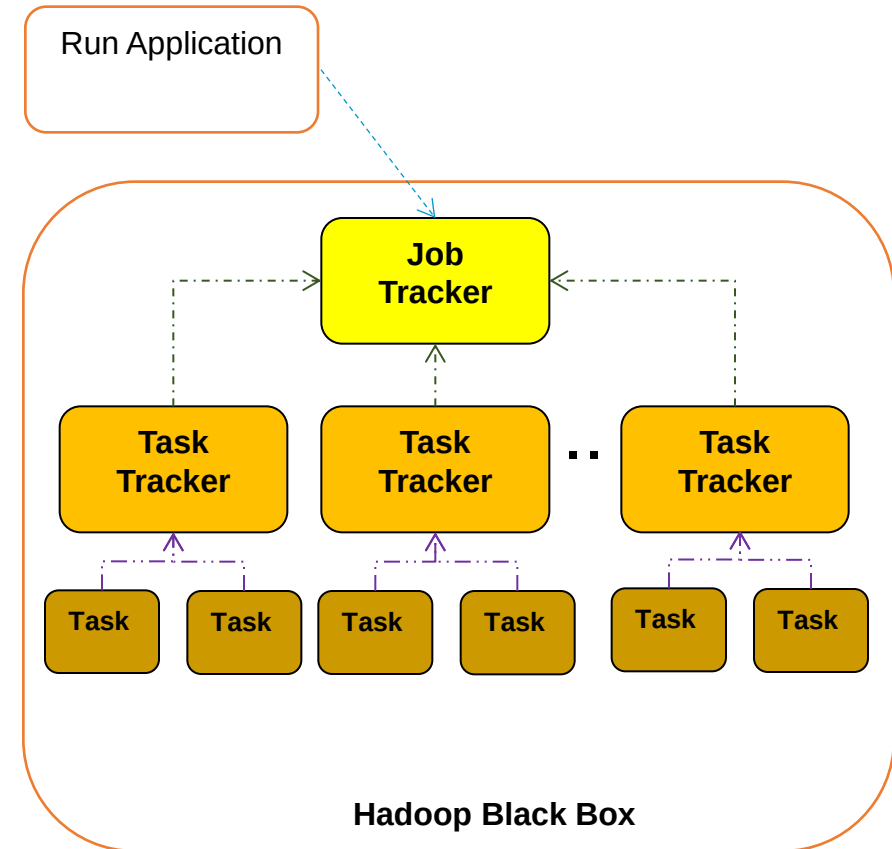
- ◆ A File is composed of blocks of size 128MB
- ◆ Blocks are duplicated on multiple data nodes (usually 3 copies)
- ◆ One name-node stores metadata (file names, block locations, etc). Usually dualize name-node.
- ◆ Optimized for huge files and sequential read
- ◆ Append-only



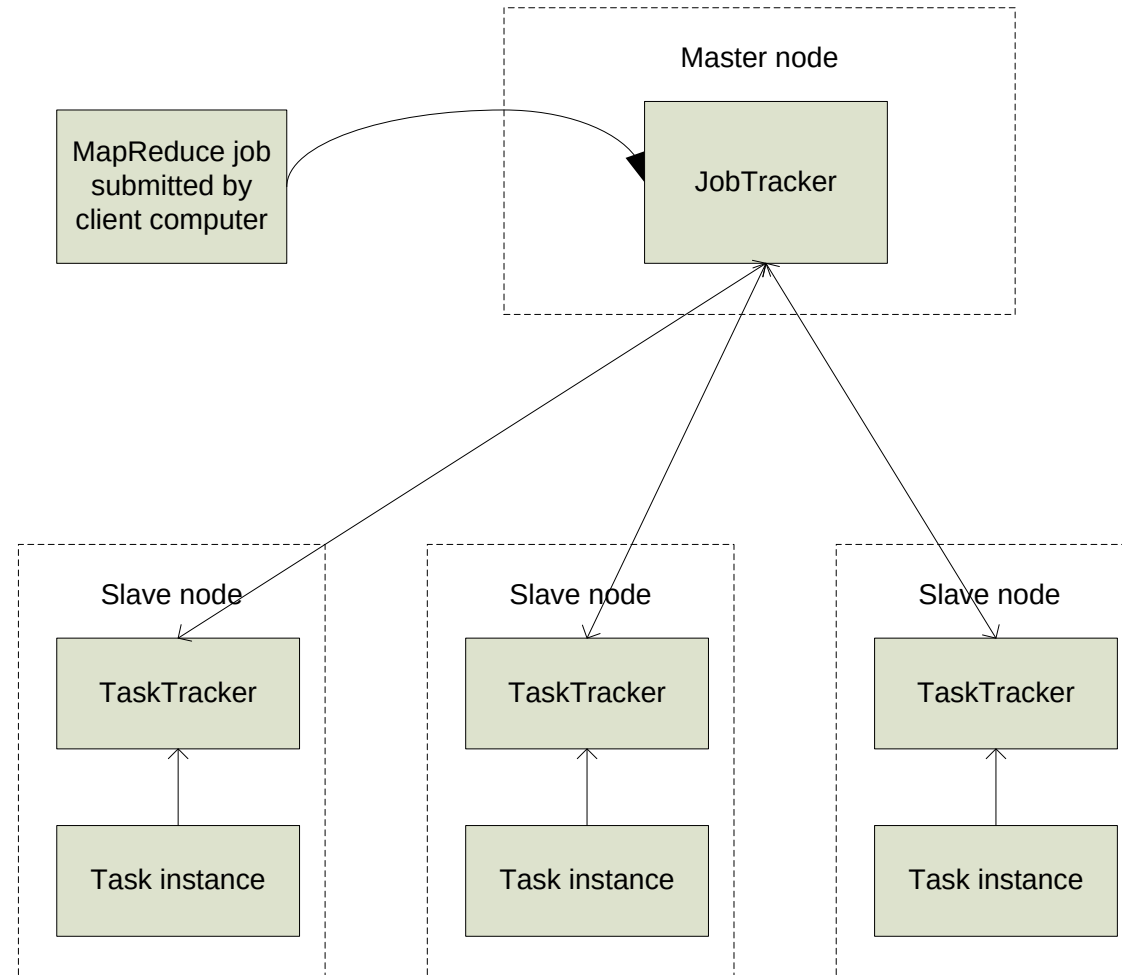
Hadoop

Principle of Hadoop's Map/Reduce Execution

- ◆ One Master controls slave
- ◆ A Mapper is placed on a node or a rack which contains the input blocks it needs to process to minimize network traffics
- ◆ When a Mapper is done with its execution, it saves the result on a local disk before sending it out to reducers.
 - When a reducer is broken, the job can be restored.
- ◆ When a task is broken, try run the task on another node.
- ◆ When a node is broken:
 - Assign other nodes to rerun tasks on the broken node.
 - Assign other nodes to rerun all the map functions run by the broken node.
- ◆ When a node is running slower than others:
 - Assign another node to run the same task and take the result coming faster. ("speculative execution")



MapReduce: High Level



Job Distribution

- MapReduce programs are contained in a Java “jar” file + an XML file containing serialized program configuration options
- Running a MapReduce job places these files into the HDFS and notifies TaskTrackers where to retrieve the relevant program code
- ... Where's the data distribution?

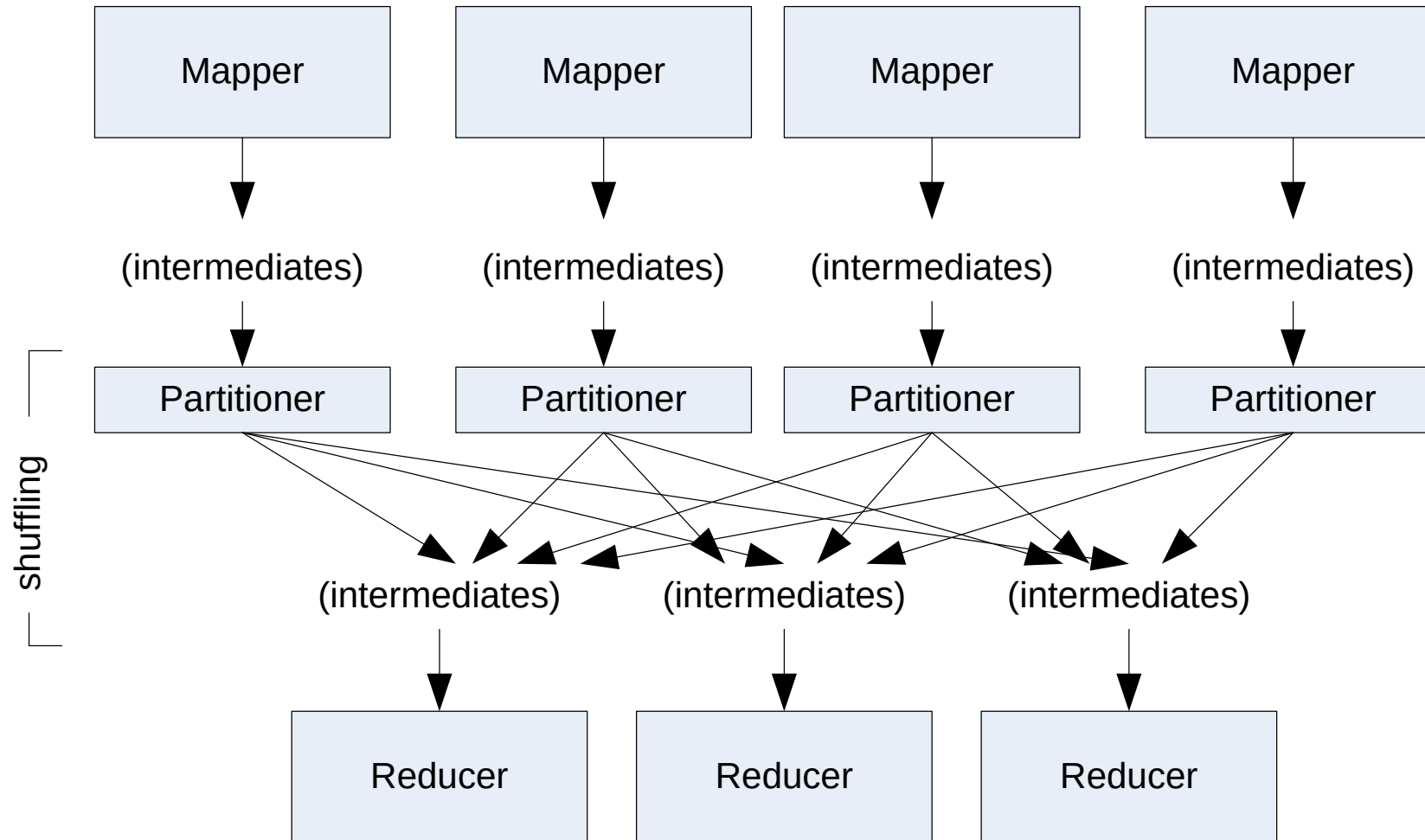
Data Distribution

- Implicit in design of MapReduce!
 - All mappers are equivalent; so map whatever data is local to a particular node in HDFS
- If lots of data does happen to pile up on the same node, nearby nodes will map instead
 - Data transfer is handled implicitly by HDFS

Reporter

- A facility for Map-Reduce applications to report progress and update counters, status information.
- Mapper and Reducer can use the Reporter provided to report progress or just indicate that they are alive.

Partition And Shuffle



Partitioner

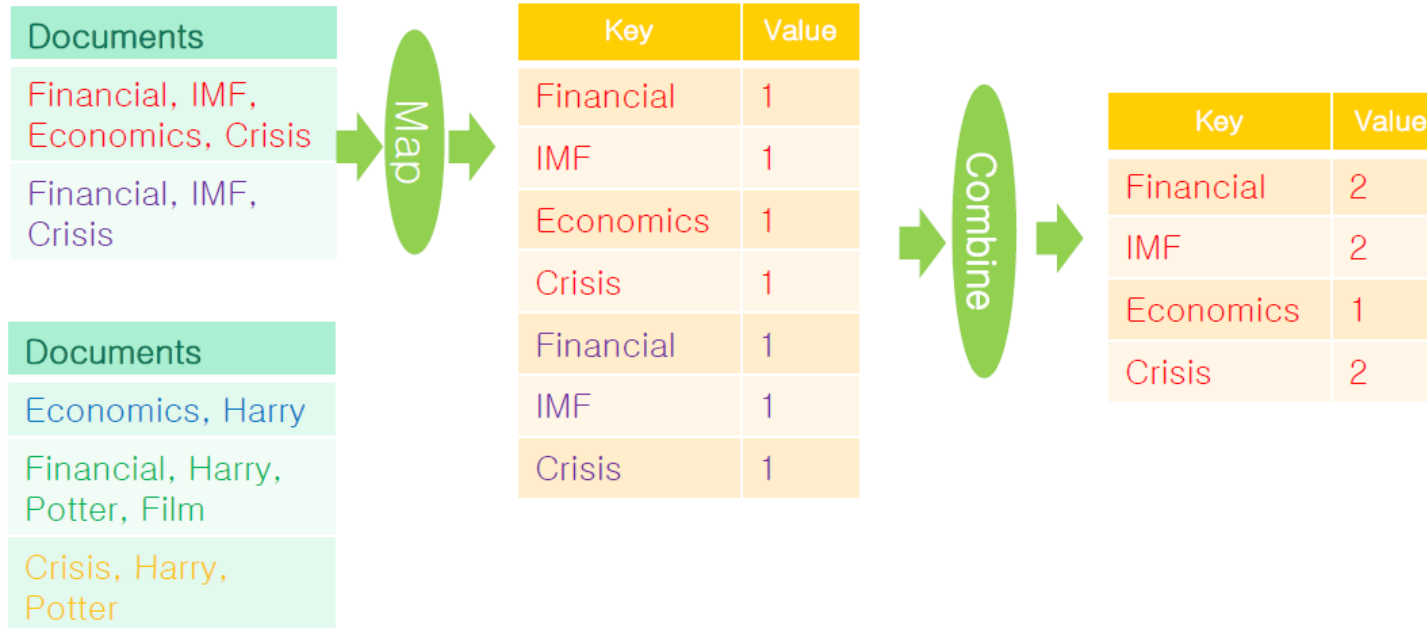
- `int getPartition(key, val, numPartitions)`
 - Outputs the partition number for a given key
 - One partition == values sent to one Reduce task
- *HashPartitioner* used by default
 - Uses `key.hashCode()` to return partition num



Combine Function

- Reduce the result size of map functions
- Perform reduce-like function in each machine
- Decrease the shuffling cost
- It is desirable to design MapReduce algorithms to use combine functions

An Example of Word Counting with Combine Function



An Example of Word Counting with Combine Function



Documents
Financial, IMF, Economics, Crisis
Financial, IMF, Crisis

Documents
Economics, Harry
Financial, Harry, Potter, Film
Crisis, Harry, Potter

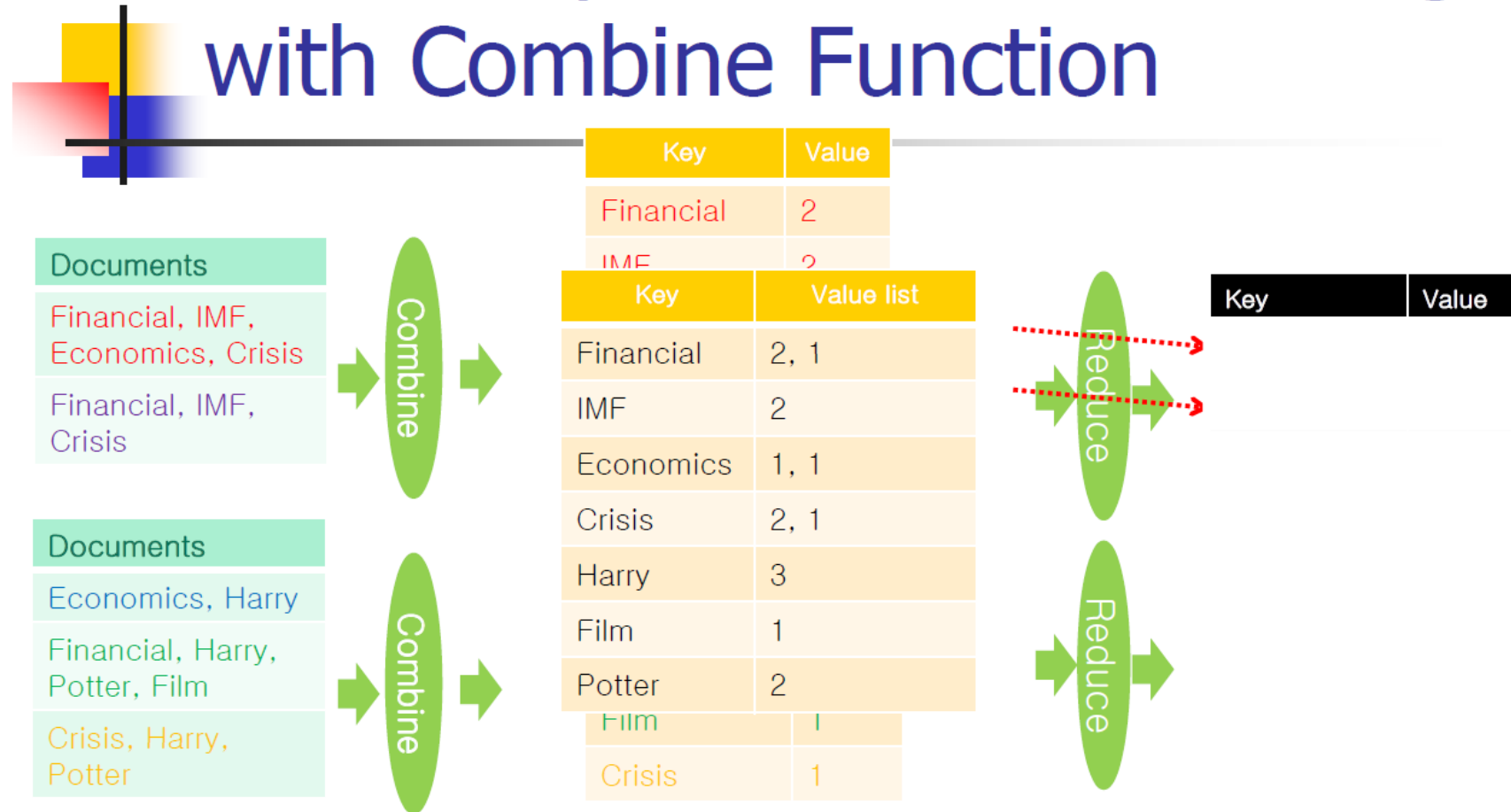


Key	Value
Economics	1
Harry	1
Financial	1
Harry	1
Potter	1
Film	1
Crisis	1
Harry	1
Potter	1



Key	Value
Economics	1
Harry	3
Financial	1
Potter	2
Film	1
Crisis	1

An Example of Word Counting with Combine Function



Before reduce functions are called,
for each distinct key, the list of its values are generated



Clustering using MapReduce

Customer1: {0, 1, 0, 1}

Customer2: {1, 0, 1, 0}

Customer3: {1, 1, 0, 0}

Customer4: {1, 1, 0, 1}

$$\text{Jaccard}(C_i, C_j) = \frac{|C_i \cap C_j|}{|C_i \cup C_j|}$$

Customer1: {0, 1, 0, 1}

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Movie1: {0, 1, 1, 1}

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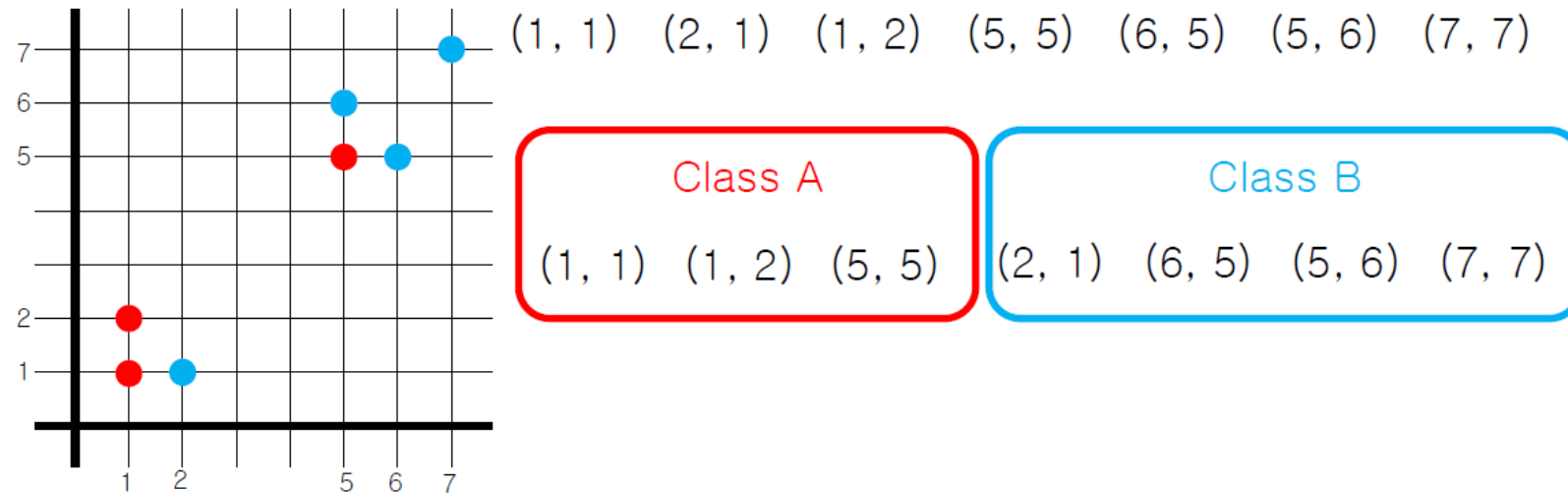
Movie4: {1, 0, 0, 1}

$$\text{Jaccard}(C_i, C_j) = \frac{|C_i \cap C_j|}{|C_i \cup C_j|}$$

K-Means Clustering Method: Partitioning Approach

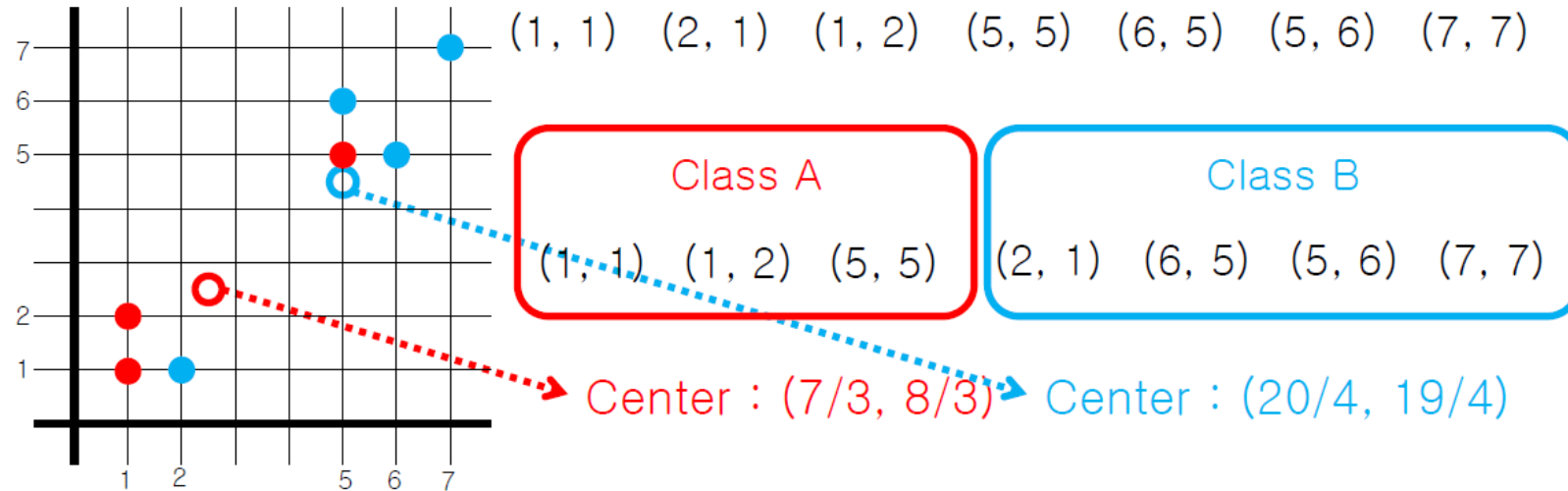
- Given k , the *k-means* algorithm performs the following repeatedly
 1. Partition objects into k nonempty subsets
 2. Compute the centroids of the clusters in the current partition (the centroid is the center, i.e., *mean point*, of the cluster)
 3. Assign each object to the cluster with the nearest centroid
 4. Stop when no more new assignments. Otherwise go back to Step 2
- The above loop finds a clustering. Thus, repeat the above many times and select the best clustering

An Illustration of K-Means Clustering ($K = 2$)

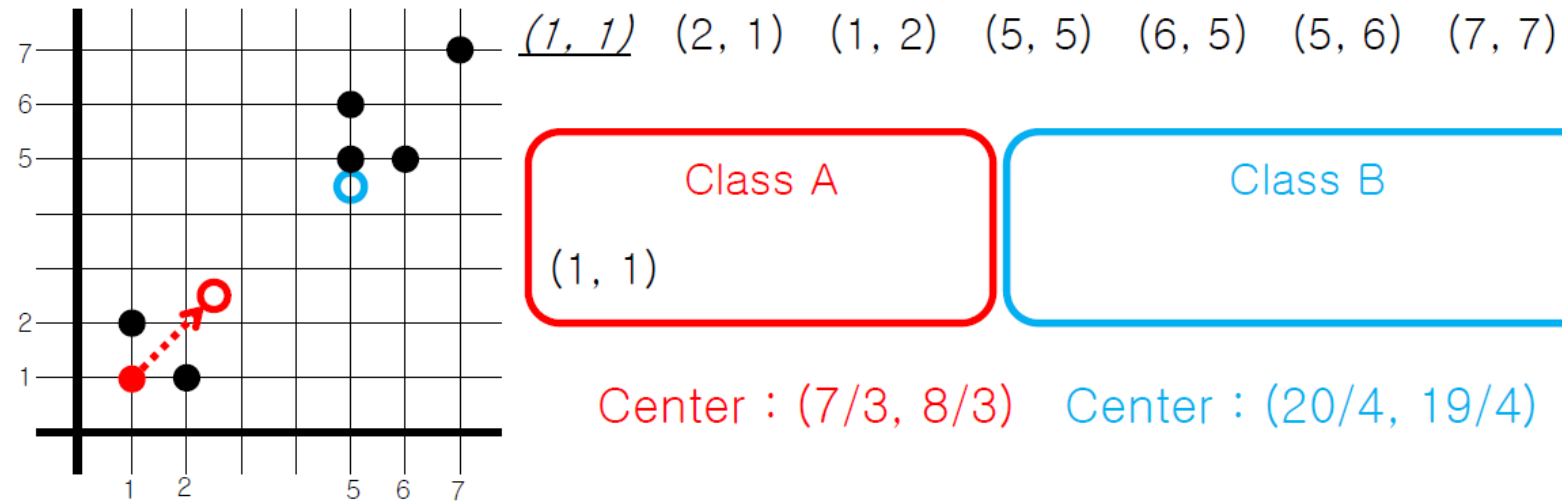


Assume we randomly partitioned the objects into 2 nonempty subsets as above!

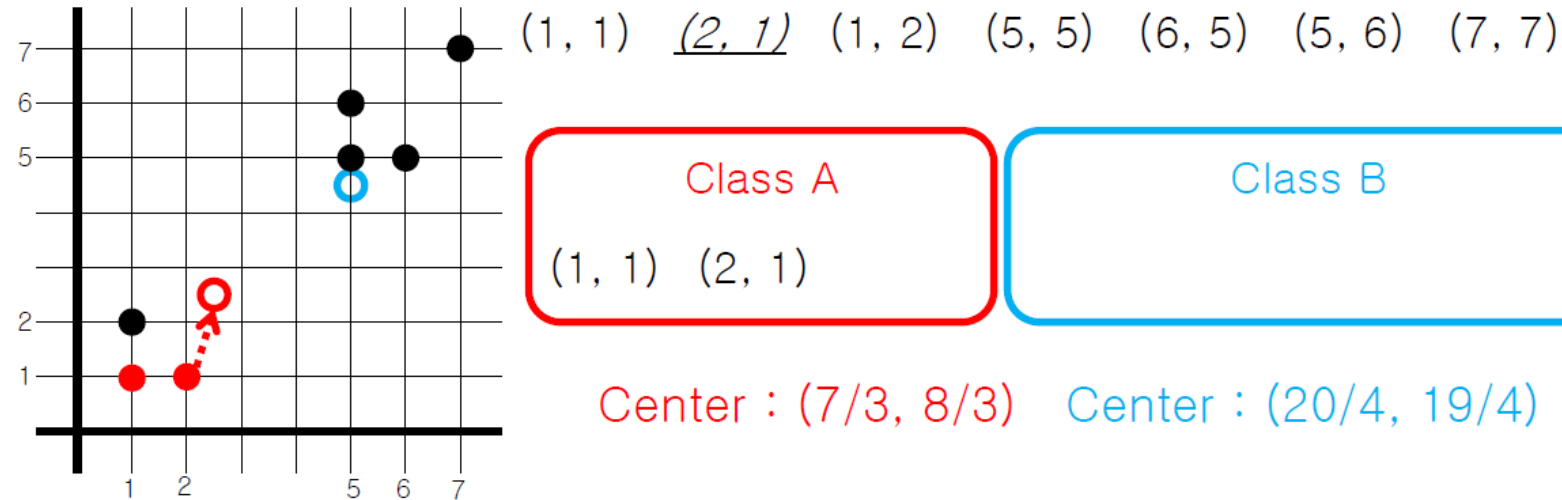
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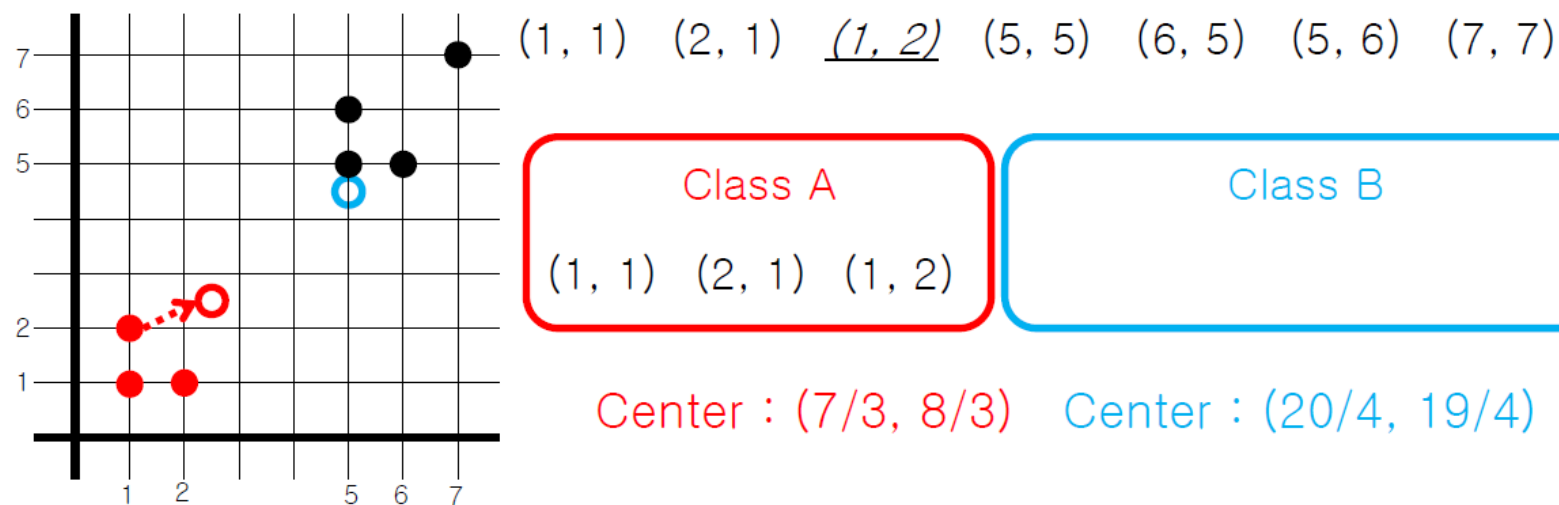
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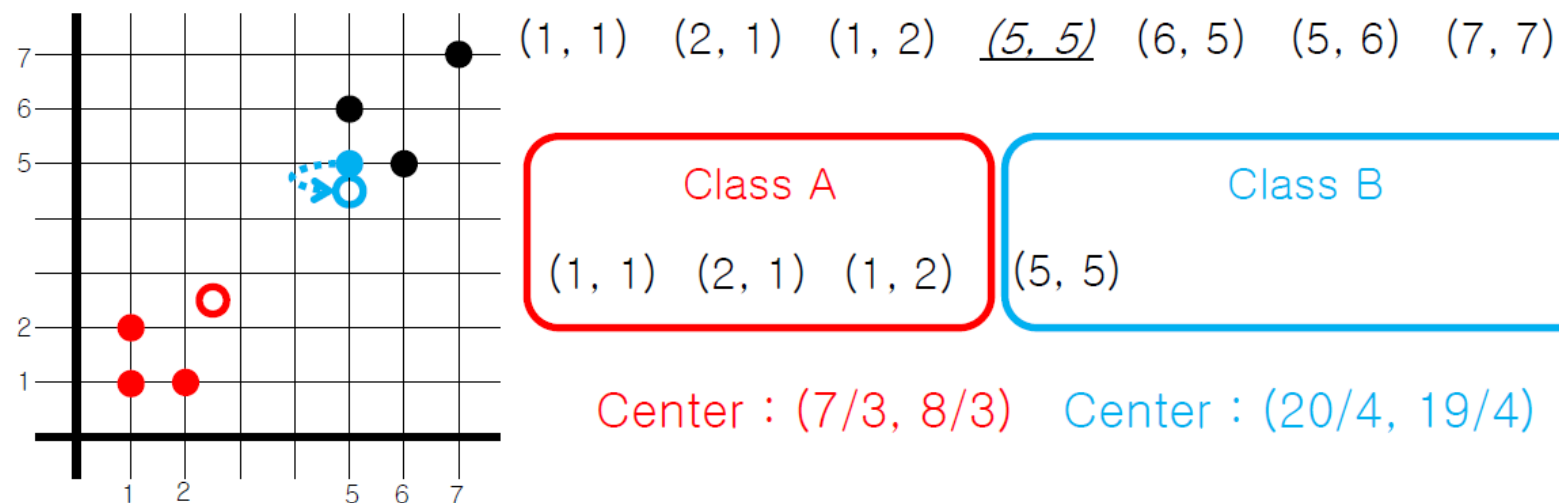
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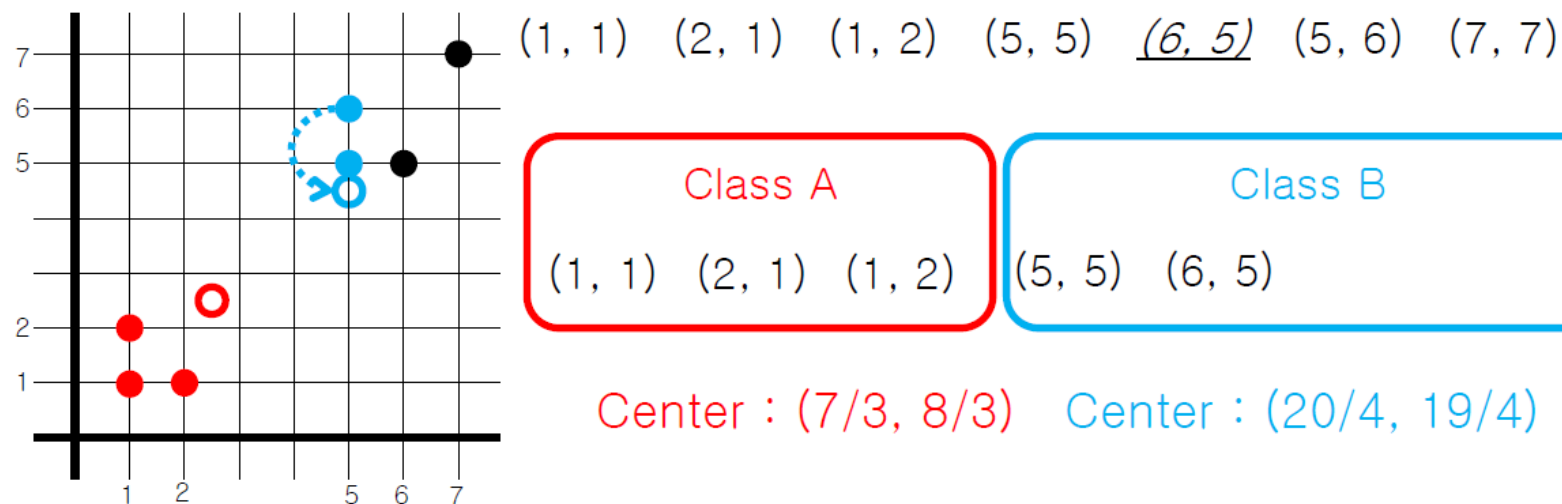
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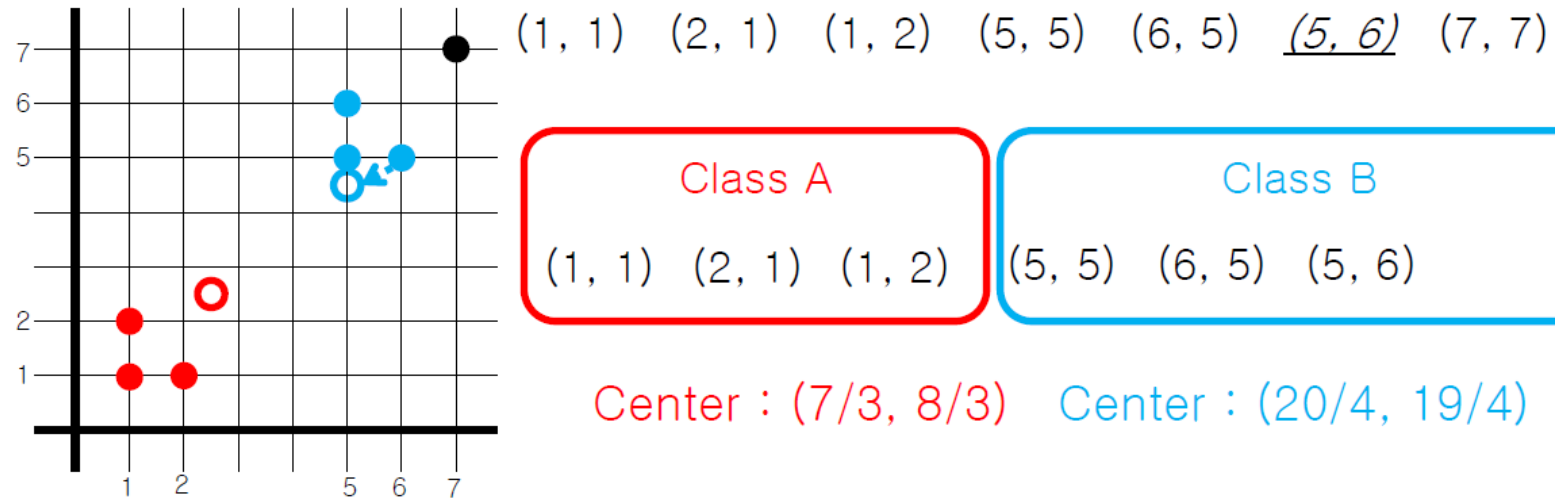
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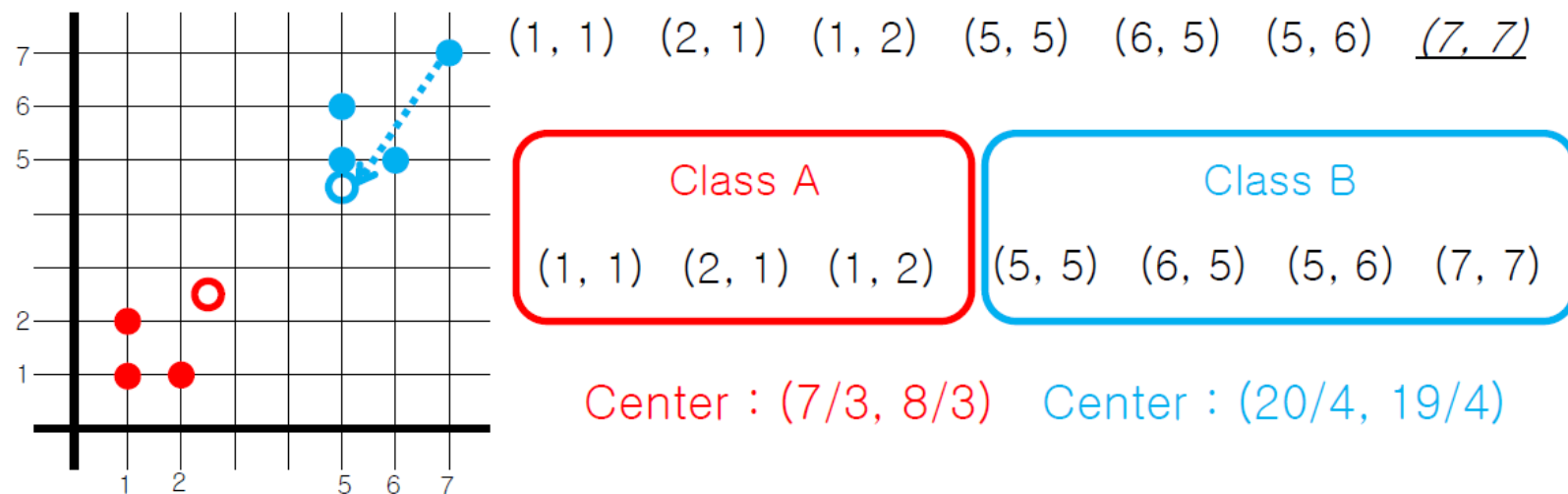
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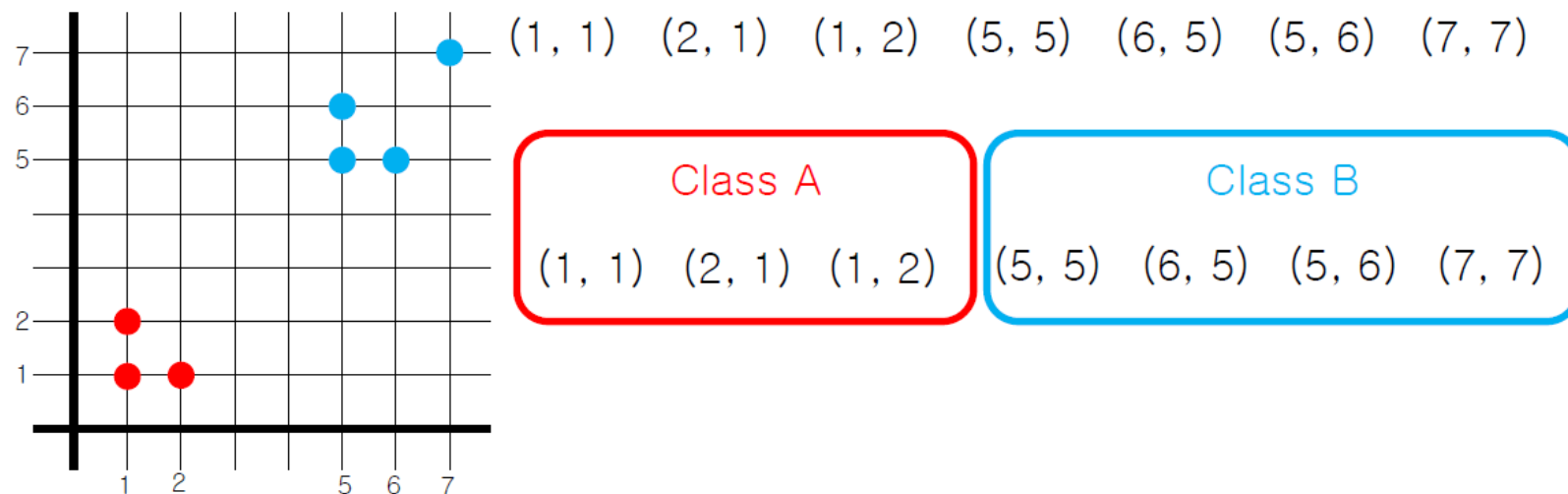
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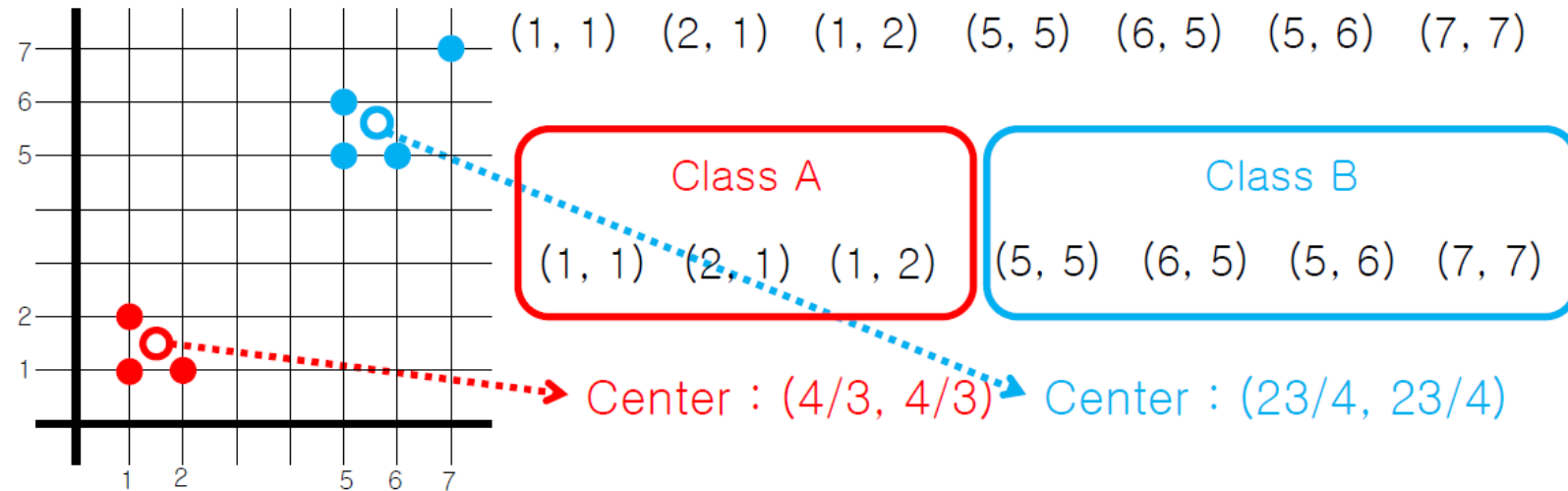
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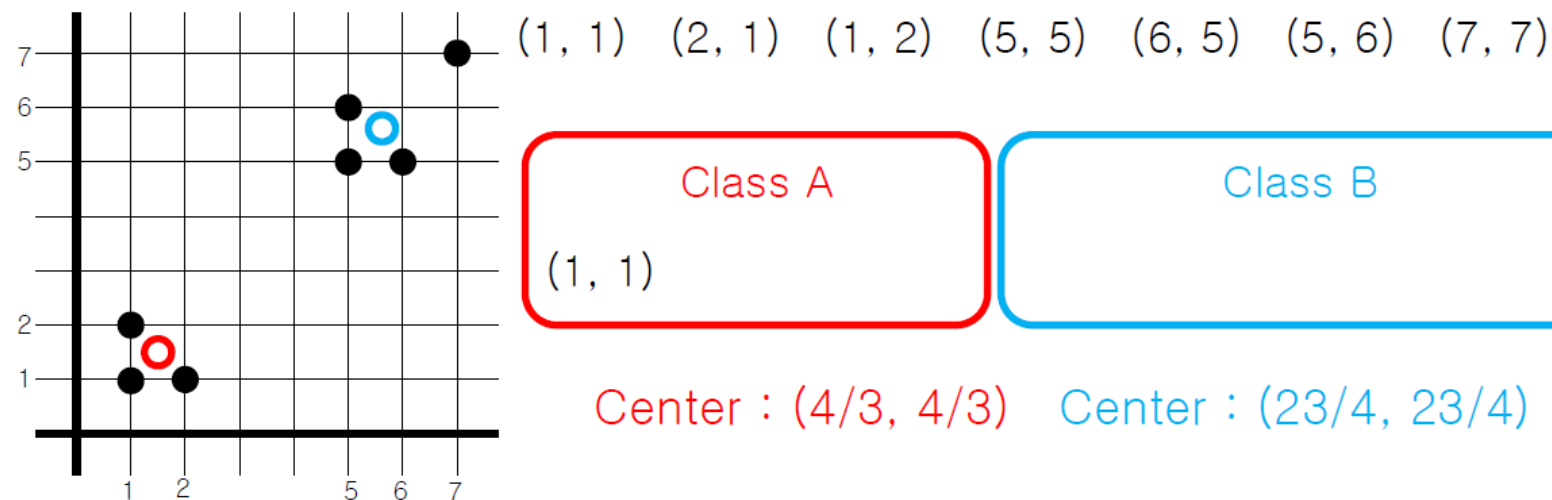


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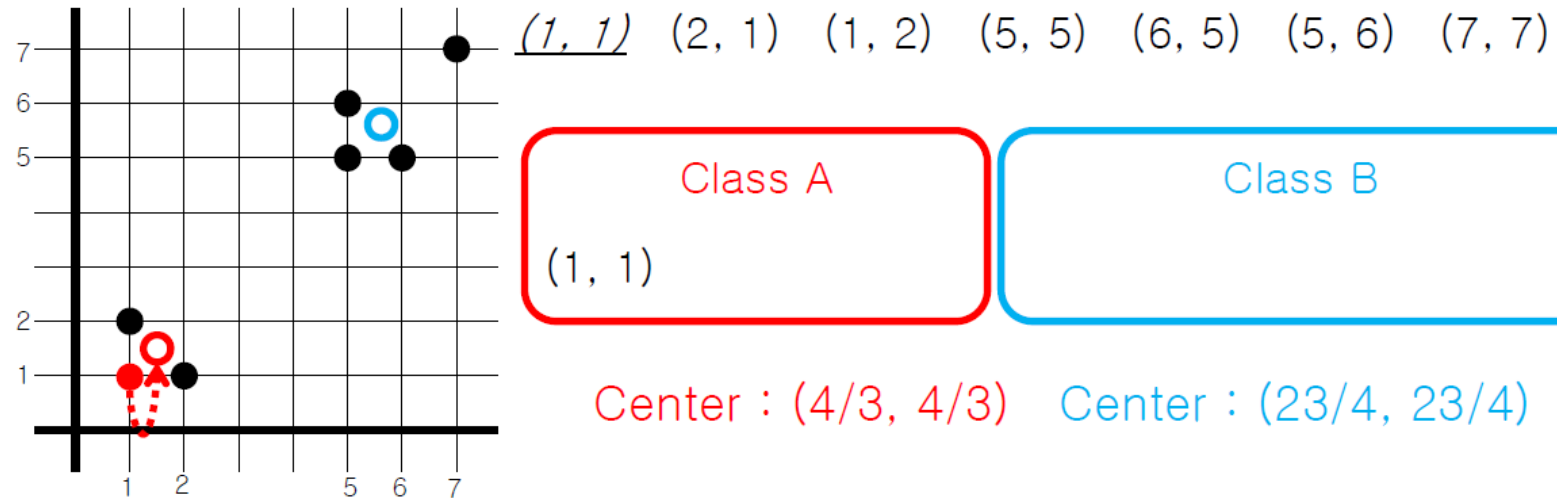


Update the cluster means

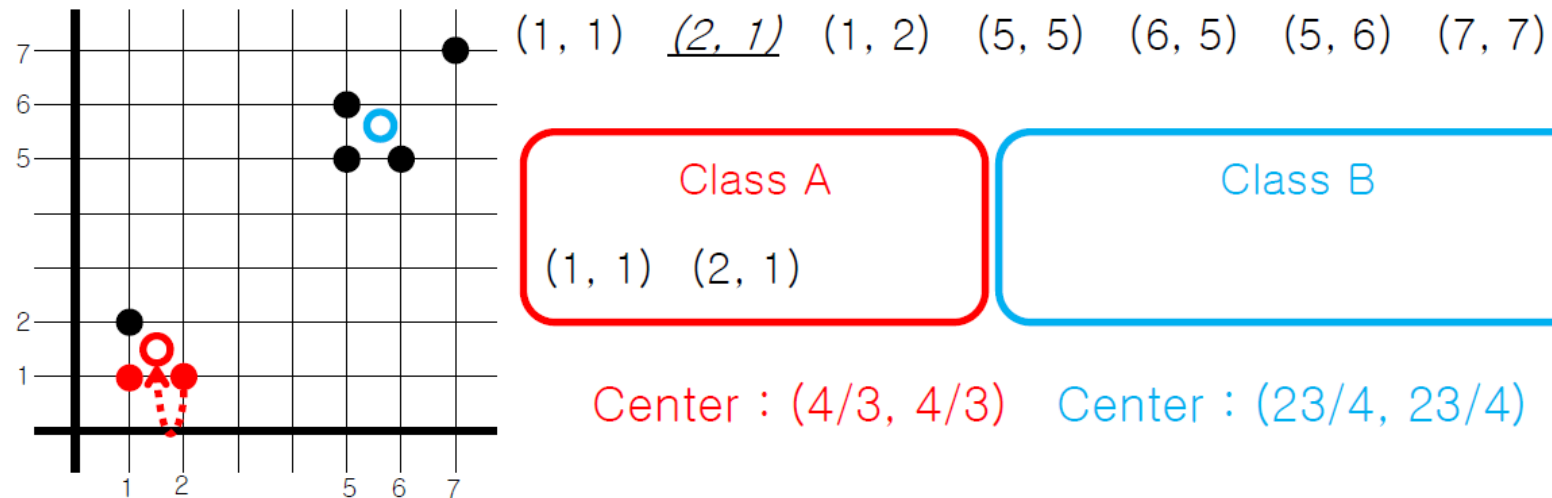
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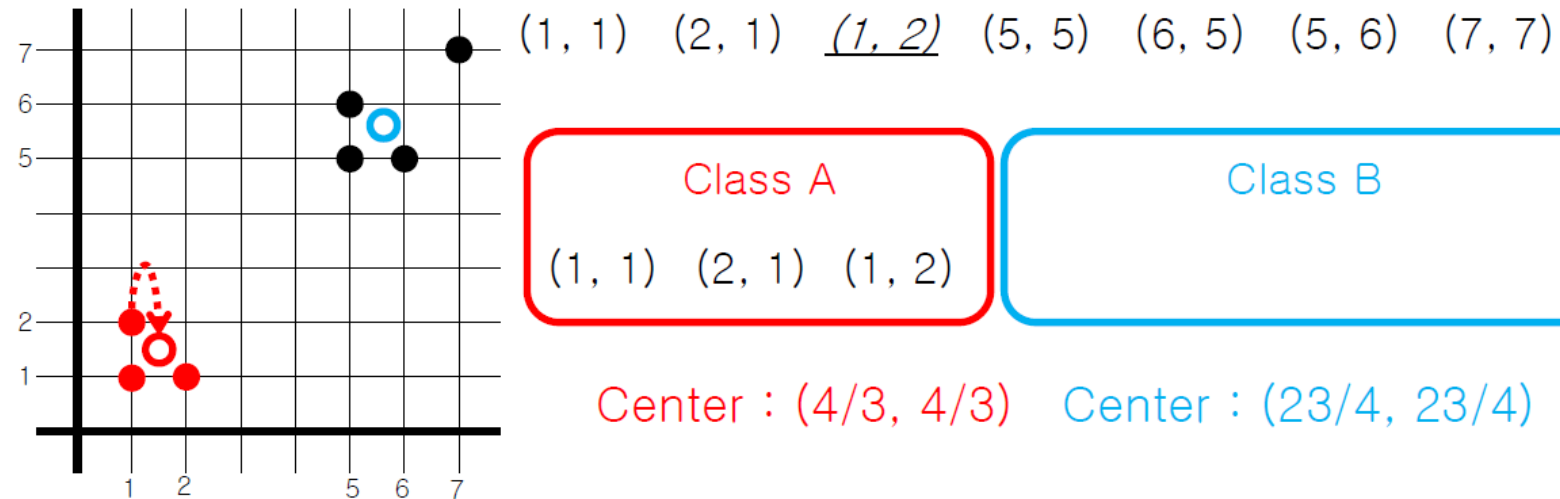
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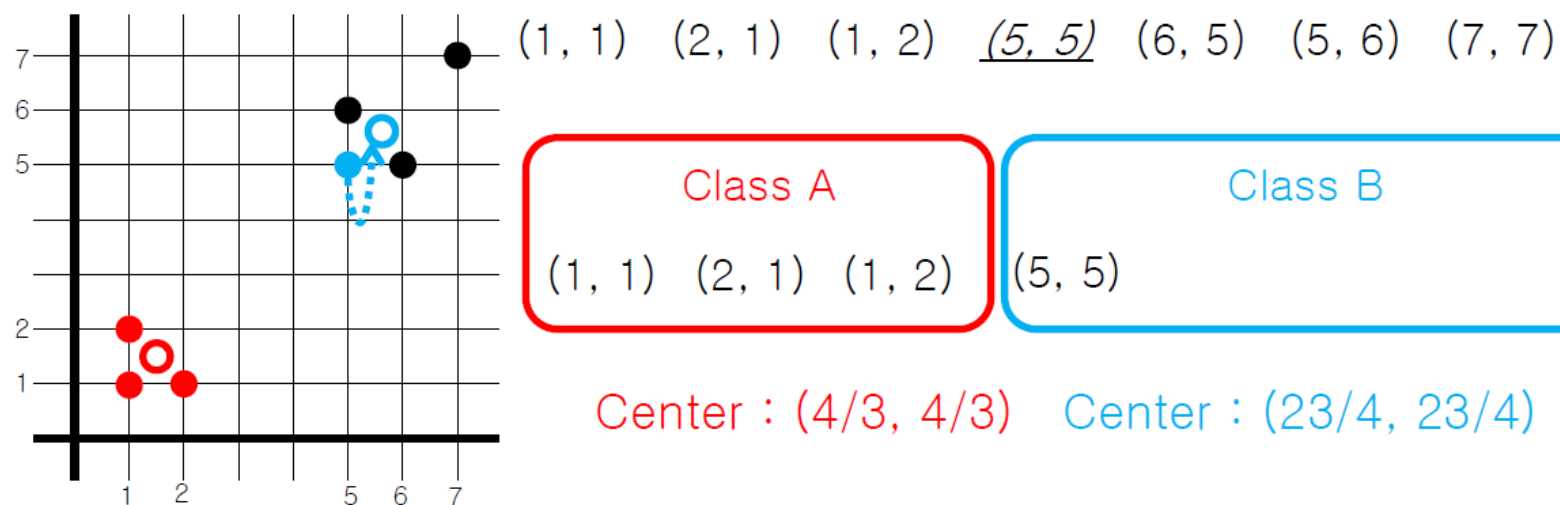
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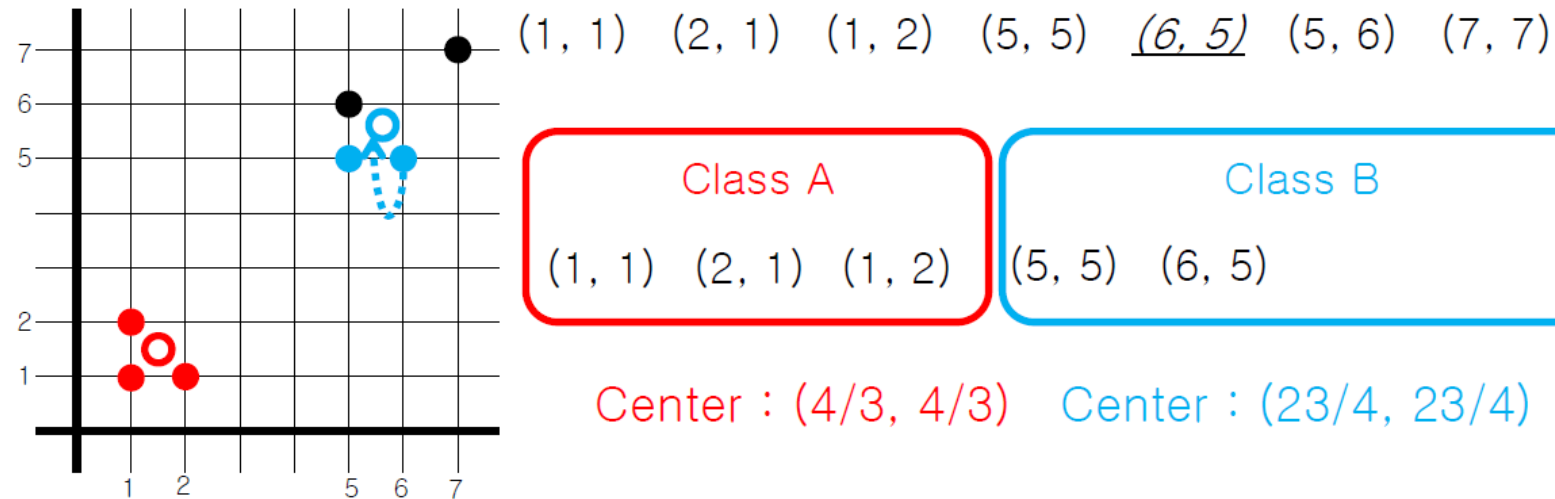
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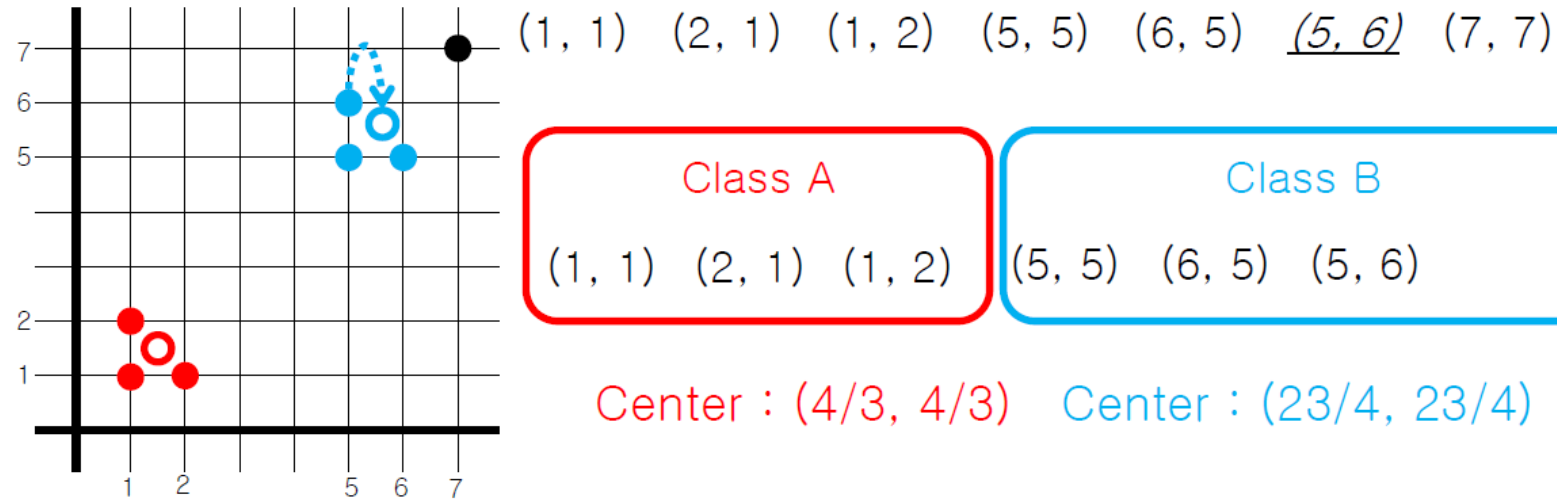
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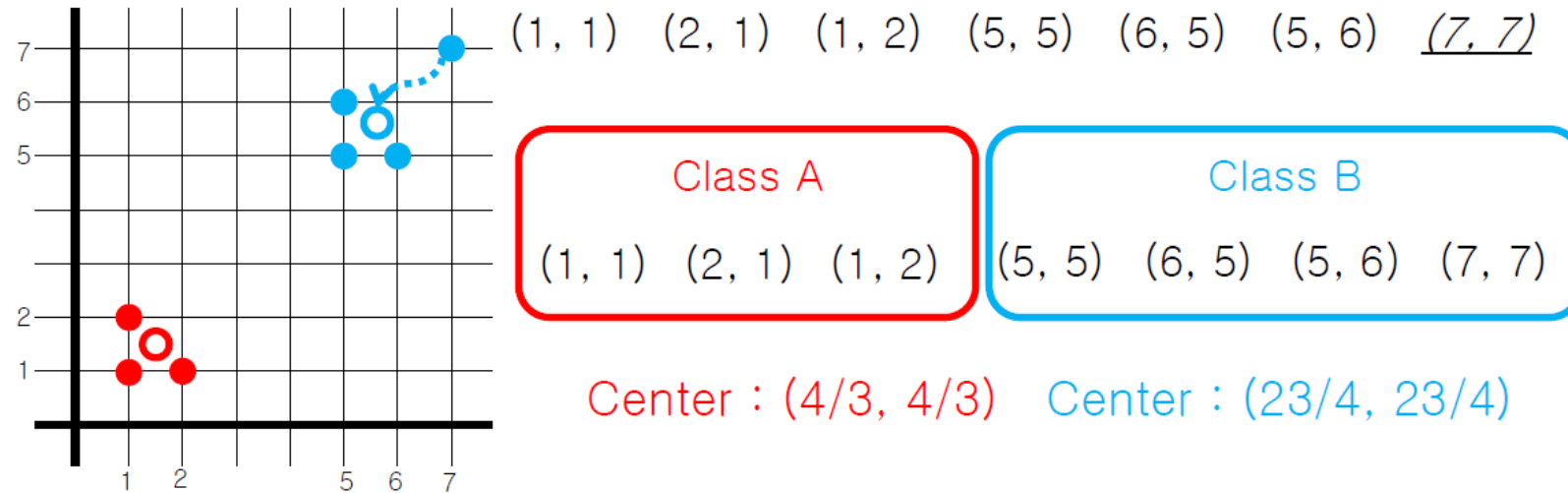
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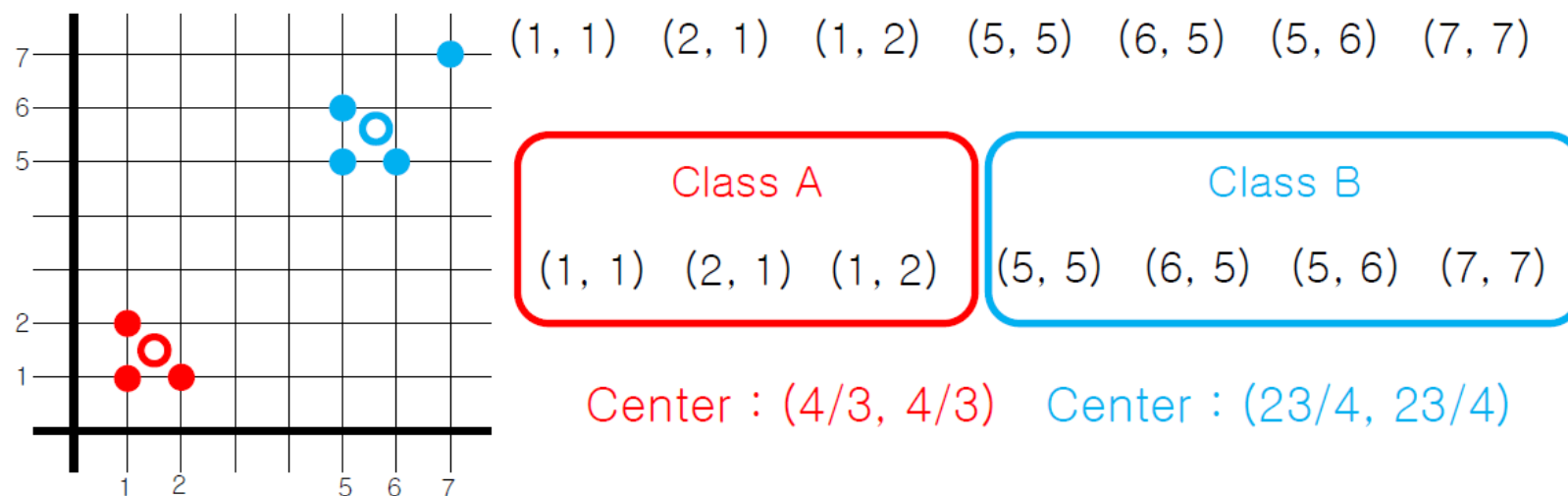
An Illustration of K-Means Clustering ($K = 2$)



An Illustration of K-Means Clustering ($K = 2$)



An Illustration of K-Means Clustering ($K = 2$)



Center : $(4/3, 4/3)$ Center : $(23/4, 23/4)$

No change $\Rightarrow$ STOP

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- The above loop finds a clustering. Thus, repeat the above many times and select the best clustering

Performance Metrics of Clustering

- Silhouette Score
 - $(n - i) / \max(n, i)$
 - n: mean nearest-cluster distance
 - i: mean intra-cluster distance